SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	B.Dhanush	**Reg. No.:**	1924260023
Date:	8/10/2008	**Duration:**	60 minutes

Code of Conduct

I B .Dhanush (1924260023)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *B .Dhanush*

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**

- Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization &	Product backlog is logically prioritized,	Backlog prioritization and sprint planning are	Prioritization or sprint planning is	Backlog prioritization and

Sprint Planning (15%)	sprint backlog is realistic, and task breakdown is complete.	mostly appropriate with minor issues.	partially complete with limited justification.	sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

1. Project Vision and Objectives

1.1 Project Vision

The Online Emergency Vehicle Routing System is a web-based platform intended to help ambulances, fire engines, and rescue vehicles reach emergency locations quickly and safely. It accepts an emergency request, records the location and emergency type, checks available vehicles, assigns a suitable vehicle, and provides a suggested route. The system aims to reduce response delays and improve coordination between users, drivers, and administrators.

1.2 Objectives

Allow users to submit an emergency request with location and emergency type.
Identify available emergency vehicles and support suitable vehicle assignment.
Provide a route from the assigned vehicle to the emergency location.
Allow drivers to receive requests and update their status.
Enable administrators to manage users, drivers, vehicles, and requests.
Keep request and vehicle information organized in one system.

1.3 Key Stakeholders and Responsibilities

Stakeholder	Role / Responsibility	Main Need
Person reporting an emergency	Submits the emergency type, location, destination and contact details.	Simple request process and status updates.
Emergency vehicle driver	Receives assigned requests, views location/route and updates vehicle status.	Clear request details and route information.
Dispatcher / emergency operator	Reviews requests and coordinates vehicle allocation when required.	Accurate, current request and vehicle information.
System administrator	Manages users, drivers, vehicles, system information and reports.	Secure management and monitoring tools.

Emergency service organization	Provides and maintains vehicles and service availability.	Better coordination and reduced response delays.
Development and maintenance team	Builds, tests, secures and maintains the application.	Clear requirements and modular design.

2. Epics and User Stories

The requirements are grouped into epics. Each user story follows the Agile format: "As a user, I want a feature, so that I receive a benefit."

Epic ID / Epic	Story ID	User Story	Priority
E1 – Emergency Request Management	US1	As a person reporting an emergency, I want to submit the emergency type, location and contact details, so that the service receives the information needed to respond.	Must
E1 – Emergency Request Management	US2	As a person reporting an emergency, I want to see the status of my request, so that I know whether it has been received and assigned.	Should
E2 – Vehicle Availability and Assignment	US3	As a dispatcher, I want to view available emergency vehicles, so that I can identify a suitable vehicle for a request.	Must
E2 – Vehicle Availability and Assignment	US4	As a dispatcher, I want to assign an available vehicle to an emergency request, so that the driver can respond to the incident.	Must
E3 – Route and Location	US5	As a driver, I want to view the emergency location and a suggested route, so that I can travel to the incident efficiently.	Must

E4 – Driver Operations	US6	As a driver, I want to update the vehicle and request status, so that the service team can monitor progress.	Should
E5 – Administration	US7	As an administrator, I want to manage users, drivers and vehicle records, so that system information remains organized.	Should
E5 – Administration	US8	As an administrator, I want to view emergency requests and generate reports, so that I can monitor system activity.	Could

3. Acceptance Criteria

Acceptance criteria are written in Given–When–Then form so each story can be checked during testing.

US1 – Submit emergency request

Given the user has opened the request form, when required fields and a valid location are entered and submitted, then the system saves the request and displays a request reference.

Given a required field is missing, when the user submits the form, then the system identifies the missing information and does not save an incomplete request.

US2 – View request status

Given a request has been saved, when the user opens its status page, then the latest recorded status is displayed.

Given the request has not yet been assigned, when its status is viewed, then it is shown as pending or awaiting assignment.

US3 – View available vehicles

Given vehicle records exist, when the dispatcher opens the vehicle list, then each vehicle's type and availability status are displayed.

Given a vehicle is marked unavailable, when the dispatcher filters available vehicles, then that vehicle is excluded from the available list.

US4 – Assign a vehicle

Given a request is pending and a vehicle is available, when the dispatcher assigns that vehicle, then the request records the assigned vehicle and the vehicle status changes to assigned or busy.

Given a vehicle is unavailable, when an assignment is attempted, then the system prevents the assignment and shows a clear message.

US5 – View suggested route

Given a vehicle and emergency location are available, when the driver opens the assigned request, then the emergency location and available route information are displayed.

Given route information cannot be loaded, when the driver opens the request, then the system informs the driver that route information is unavailable.

US6 – Update status

Given the driver is logged in and has an assigned request, when the driver selects a valid status, then the system saves and displays the updated status.

Given the request is completed, when the driver marks it completed, then the request is shown as completed in the system.

US7 – Manage records

Given the administrator is authenticated, when the administrator adds or updates a vehicle or driver record with valid data, then the system saves the change.

Given a user without administrator permission attempts to open administration functions, then access is denied.

US8 – View reports

Given emergency request records exist, when the administrator opens the report view, then the system displays a summary of recorded requests and statuses.

Given no records match the selected report criteria, then the system displays an empty-result message rather than an error.

4. Product Backlog Prioritization

The backlog is prioritized using MoSCoW: Must have, Should have, and Could have. Emergency request intake, vehicle availability, assignment and routing are prioritized because they form the core workflow.

Rank	Story	Priority	Reason / Dependency
1	US1 – Submit emergency request	Must	Core entry point; required before assignment.
2	US3 – View available vehicles	Must	Required to identify a suitable vehicle.
3	US4 – Assign a vehicle	Must	Depends on saved requests and vehicle availability.
4	US5 – View suggested route	Must	Needed by the driver after assignment.
5	US6 – Update status	Should	Improves coordination and request tracking.
6	US2 – View request status	Should	Keeps the person reporting informed.
7	US7 – Manage records	Should	Supports ongoing system administration.
8	US8 – View reports	Could	Useful for monitoring, but not essential to the first working flow.

5. Sprint Backlog – Sprint 1

Sprint duration: 2 weeks (proposed). The first sprint focuses on a basic end-to-end emergency request and vehicle assignment workflow. Route integration is planned as a simple route/location view or initial route service, depending on available map integration.

5.1 Selected User Stories and Development Tasks

Story	Tasks	Deliverable

US1	Design request form; validate emergency type, location and contact details; save request; show confirmation/reference.	Emergency request submission works and stores a request.
US3	Create vehicle data structure; add vehicle type and availability fields; build available-vehicle list.	Dispatcher can view available vehicles.
US4	Build assignment action; validate vehicle availability; link vehicle to request; update statuses.	Dispatcher can assign an available vehicle to a request.
US5 (basic)	Display the saved emergency location; connect or prepare the route view; show a message if route data is unavailable.	Driver can open the assigned request and see its location/route information.

5.2 Sprint Task Checklist

Confirm the request, vehicle and assignment data fields.

Create the database tables and basic application pages.

Implement request submission and server-side validation.

Implement the available-vehicle list.

Implement assignment rules and status updates.

Build the driver view for assigned requests and location/route information.

Test valid requests, missing fields, unavailable vehicles and assignment status changes.

Review the working increment and record defects for the next sprint.

6. Story Point Estimation

Estimation technique: Planning Poker using the Fibonacci sequence (1, 2, 3, 5, 8, 13). Points reflect relative effort, complexity, uncertainty and testing needs; they are not hours.

Story	Points	Justification

US1 – Submit emergency request	5	Form, field validation, location/contact data, persistence and confirmation.
US3 – View available vehicles	3	Vehicle list and availability filtering are moderate in scope.
US4 – Assign a vehicle	5	Requires request–vehicle linking, availability checks and status updates.
US5 – View suggested route (basic)	8	Location presentation and route-service integration have greater uncertainty.
Total planned	21	Provisional estimate for the sprint; adjust after team capacity and technical review.

Planning assumption: the team has enough capacity for approximately 21 points in this proposed sprint. Actual velocity should be measured after the first sprint and used to plan later sprints.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

Enable a dispatcher to receive a saved emergency request, view available vehicles, assign a suitable available vehicle, and allow the driver to view the assigned request and its emergency location.

7.2 Expected Deliverables

A working emergency request form with basic validation and confirmation.

Stored emergency request records.

A vehicle list showing vehicle type and availability.

Vehicle assignment with checks that prevent assigning an unavailable vehicle.

Updated request and vehicle status after assignment.

A basic driver page showing assigned request and emergency location/route information.

Test results for the main workflow and common validation failures.

8. Definition of Done and Review

Each selected story meets its acceptance criteria.

Code is integrated and reviewed by the team.

Input validation and access checks are implemented where relevant.

Core workflow tests pass, including unavailable-vehicle handling.

No known critical defect prevents the sprint demonstration.

The working increment is demonstrated to the product owner/stakeholders and feedback is added to the backlog.

9. Risks, Assumptions and Future Sprint Items

9.1 Risks and Assumptions

Accurate location and route guidance depend on the availability and reliability of mapping/GPS services.

Vehicle availability must be updated correctly; outdated records may lead to unsuitable assignment.

Emergency and contact information must be protected with authentication and access control.

This sprint plan assumes a two-week sprint and a team capacity of about 21 story points; these are planning assumptions, not measured results.

The system supports coordination but does not replace emergency operators or official emergency-service procedures.

9.2 Future Sprint Items

User-facing request status tracking and more detailed notifications.

Driver status updates and completion workflow.

Administrator tools for managing users, drivers and vehicles.

Real-time GPS tracking and traffic-aware route calculation, where supported by external services.

Reports, estimated time of arrival, mobile access and possible AI-based route optimization.

10. Conclusion

This Agile sprint plan translates the Online Emergency Vehicle Routing System case study into a prioritized product backlog, testable user stories, acceptance

criteria, and a proposed first sprint. The first sprint delivers the foundation of the emergency workflow: submitting a request, viewing available vehicles, assigning a vehicle and displaying the emergency location to the driver. Feedback from the sprint review will be used to refine priorities and plan subsequent increments.